Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/11/2014
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Mapping Spaceflight-Induced Hypoxic Signaling And Response		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 6/11/2015 per PI and NSSC information (Ed., 6/20/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	This research capitalizes on a rapid turn-around spaceflight opportunity and the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFU) hardware to address how spaceflight affects gene expression related to hypoxic response in Arabidopsis. Root zone hypoxia is thought develop in spaceflight as weightlessness leads to a reduction in the buoyancy-driven convection that usually aids in gas exchange around organisms. This in turn leads to the development of oxygen-limiting conditions with adverse effects on plant vigor and yields. Our analysis of Ca2+ signaling responses and associated gene expression patterns in Arabidopsis indicates Ca2+ signaling is linked to anoxic response in plants. We have identified a suite of Ca2+-dependent proteins that show rapid transcriptional responses to anoxia in both roots and shoots. Mutants in two such genes, aca1-2, and cax2-2 (both Ca2+ transporters) are resistant to anoxic challenge. These mutants also show disrupted gravitropic response suggesting that Ca2+ signaling may provide a point of cross-talk between the anoxic and gravity response systems. We therefore propose to compare growth and gene expression patterns in aca1-2, cax2-2, and wild-type plants. Seedlings will be grown on orbit and fixed in RNAlater for subsequent analysis. Differences in growth will be quantified and correlated with transcription patterns analyzed post-flight by a combination of microarray and QPCR analyses. Comparison to ground-based controls maintained under normal and varying degrees of anoxic conditions at 1xG, or during clinorotation, will be used to ask how much of the responses seen on orbit can be explained by developing anoxia, disruption of gravisensing or a combination of anoxic challenge against the background of microgravity. Results from this analysis are expected to not only help advance our understanding of the development of anoxic response in space and on the Earth but also test whether the Ca2+ signaling system provides targets for genetically engineering potential countermeasures to this stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how low oxygen sensing operates in plants, targeting the Ca2+-dependent signaling machinery of the plant root. The results from this work will not only provide insight into how low oxygen levels impact plant growth in space but also has impact on understanding similar responses on Earth. The low oxygen sensing/response systems triggered by anoxic conditions in space are also those employed by plants challenged by low oxygen on Earth, such as during flooding of the soil. Thus, this work aimed at analysis of how Ca2+ signaling is involved in the low oxygen response induced during spaceflight is also revealing a role for this system in this general flooding response. These results are therefore providing molecular targets for potential manipulation to tailor plant growth to varied soil oxygen status, such as promoting flooding tolerance in crop plants.
Task Progress:	This project has completed the development, flight, and sample recovery phases and sample analysis is currently underway. Flight The experimental setup for this BRIC-17 experiment was using surface sterilized seeds of Arabidopsis plated on nutrient agar medium in 60 mm Petri dishes. The sterile plates were then loaded into Petri Dish Fixation Units (PDFU) and 5 PDFUs plus a HOHO data logger secured inside the Biological Research in Canister (BRIC) hardware. Payload verification tests showed >90% germination of these seeds in the BRIC-PDFU hardware. Germination was reliably delayed by holding the plates at 4°C until the required time for seedling growth, indicating this as a suitable approach to place un-germinated seeds onto the International Space Station. Germination was then reliably started by warming the equipment to room temperature. This configuration was successfully launched using the SpaceX2 vehicle on March 1st, 2013. The hardware consisted of two BRIC canisters, each containing five PDFUs and one HOBO temperature data logger. Plates of seeds (4 plates of wild-type and 3 each of two anoxia-related CAX2 Ca2+ transporter mutants were integrated into the PDFUs on 2/27/2013). These were then placed in a double cold stowage bag maintained at 4°C and stowed into the SpaceX-2 Dragon capsule at Launch -24 hrs. This hardware was then successfully carried to the ISS where it was de-stowed from the Dragon capsule and transferred to the US Lab. After 8 days at room temperature, the crew injected RNAlater fixative into the PDFUs and the following day, the canisters were placed in the MELFI freezer onboard the ISS at -95°C, stored for 11 days and then returned using the Dragon capsule in a -32°C Cold Bag on March 24, 2013. These samples were de-integrated from the hardware on April 4, 2013 and transported to the University of Wisconsin-Madison for further analysis Postflight analysis: Initial analysis revealed some fungal contamination on some plates (identified as penicilium) but sufficient intact samples were recovered to allow for the gene expression analyses required from this experiment. Germination efficiency was >90% on all plates. Seedling growth was assessed from digital images of the thawed seedlings and a redirection from shoot to root growth was observed in the spaceflight materials versus the ground-based controls. This reallocation of growth was missing in a cax2 mutant suggesting the mutant may be resistant to some of the stresses of spaceflight that are remodeling growth. All samples yielded sufficient quantity and quality of RNA for both quantitative PCR and whole genome transcriptional analyses to monitor gene expression. Isolated RNA was provided to the University of Wisconsin-Madison Biotech Center who performed library construction and RNAseq-based transcriptome profiling for gene expression analysis. Raw data from this RNAseq analysis was available towards the beginning of 2014 and is currently under study. Preliminary analysis indicates 38 genes are up-regulated in spaceflight in wild-type and cax2 mutant plants, implying a core set of spaceflight-related genes. These cover a range of functions from hormone response genes to those related to oxidative stress. In addition, the induction of 210 other genes was suppressed in the cax2 mutants relative to wild type, suggesting these are targets of the CAX2-dependent machinery, and so strong candidates for genes that are part of the spaceflight anoxic response that is triggered by the CAX2 protein. Current work is further analyzing these patterns of gene transcription and comparing them to patterns seen in response to experimentally-imposed low oxygen levels on Earth. Ground-based analysis In parallel to the flight experiment, wild-type, and the Ca2+ transporter mutants used in the spaceflight experiment have been monitored for their abilities to trigger cellular Ca2+ changes in response to low oxygen stress. Both cax2 mutants showed aberrant Ca2+ signals in the root tip to low oxygen, with increased magnitude of Ca2+ elevations. As a positive control for a stress that is known to induce large Ca2+ changes, salt stress was also used. Local application of salt stress

	to the root was observed to trigger a previously unknown wave of Ca2+ response that traversed the root system, traveling at 400 µm/sec. This Ca2+ wave was dependent on the vacuolar Ca2+ channel Two Pore Channel 1 (TPC1) and was required to trigger stress responses in the plant, as measured by patterns of stress-related gene transcription. Effects of mutations in cax2 on this Ca2+ wave signaling system are under way.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
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